

PHYSICAL EDUCATION MASTERY GUIDE

Class 12 CBSE (Based on Source Paper Analysis)

100% RECURRING TOPICS IDENTIFIED

1. CORE MANDATORY CATEGORIES (100% PROBABILITY)

Tournament Fixtures & Formulas

Drawing **Knock-out** and **League** (Cyclic/Staircase) fixtures for odd/even teams is certain. Calculation Focus: Next Power of 2, placement of Byes in halves, and Seeding logic.

Yoga for Lifestyle Management

Focus on 4 pillars: **Asthma, Hypertension, Obesity, and Back Pain**. You must know Procedure, Benefits, and Contraindications for specific poses like Bhujangasana or Vajrasana.

Applied Biomechanics

Newton's 3 Laws with sports examples. Identification of **Levers** (1st, 2nd, 3rd Class) and factors affecting **Equilibrium** (COG, Base of Support).

CWSN Empowerment

Founding facts, logos, and mottos for **Paralympics, Special Olympics, and Deaflympics**. Strategies for inclusive curriculum modification.

2. HIGH-WEIGHTAGE UNITS (LONG ANSWERS)

Physiology and Sports Injuries (Unit 7)

- **Identification:** Greenstick, Comminuted, Oblique fractures (often image-based).
- **Exercise Effects:** Muscular hypertrophy, Stroke volume, Cardiac output.
- **Soft Tissue:** Sprain (ligament) vs Strain (tendon/muscle).

Tests and Measurement (Unit 6)

- **Rikli & Jones:** Six-subtests for Senior Citizens (Chair Stand, Back Scratch, etc.).
- **SAI Khelo India:** Plates Tapping and Flamingo Balance (Ages 5-8).

3. QUICK-REFERENCE FORMULA CHEAT SHEET

Metric	Formula / Logic
Knock-out Matches	$N - 1$
League Matches	$[N(N-1)] / 2$
Total Byes	(Next Power of 2) - N
Harvard Step Test FI	$(100 \times \text{duration in sec}) / (2 \times \text{sum of heart beats})$
Body Mass Index (BMI)	$\text{Weight (kg)} / \text{Height (m)}^2$
Nutrition Classes	Fat Soluble [A,D,E,K] Water Soluble [B,C]

4. EXAM-DAY STRATEGY

- 1. **Study the Visuals:** 15-20% of the paper uses images. Study the diagrams of all asanas and fracture types.
- 2. **Section D Mastery:** Case studies typically target Tournament calculations or CWSN inclusion strategies.
- 3. **Differentiation:** Be ready to distinguish "Intramural vs Extramural" and "Isometric vs Isotonic" exercises.

Strategic Study Resource | Created by @drgalaxyx | Time Optimized for Board Revision